

Privacy-Preserving and Scalable Occupant Spatial Tracking in Indoor Environments

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Abstract

Spatiotemporal tracking in indoor environments faces privacy concerns and complexity that affect widespread adoption. This work presents a scalable, privacy-preserving tracking system designed to overcome these barriers. We propose a wire-free, edge-computing architecture that fuses camera and mmWave radar data locally, transmitting only anonymous identity vectors and spatial coordinates. Connectivity is achieved via a self-forming mesh network on the low-interference DECT NR+ 1.9 GHz band, ensuring scalable and low-latency communication. The proposed system will be validated through real-world deployments in Cambridge and London, and will deliver granular tracking data of occupants in buildings while strictly protecting their identities, thereby offering a practical solution for diverse indoor environments.

Introduction

Occupant spatial tracking drives critical applications in building optimisation [1] [2], energy reduction [3], shopper behaviour analysis [4][5], and optimising patient flow in clinics to reduce wait times [6] [7]. However, existing solutions have certain privacy risks which prevent people from participating or agreeing to be tracked [6]. In addition, the complex process of installing and deploying the needed infrastructure adds further hurdles to practicality and scalability.

This work aims to design an indoor tracking system to deliver spatiotemporal tracking data while being privacy-preserving, scalable, and easy to deploy.

Overview of the proposed work

We use edge computing to process data directly on the sensor units, ensuring raw sensor data never leaves the device, thereby contributing to preserving privacy. In addition, we use a wire-free mesh network architecture to address deployment hurdles.

At the network edge, we fuse camera vision and mmWave radar sensor data to: (1) extract visual features of individuals to create anonymous identity vectors, (2) use the depth data from the privacy-friendly radar to spatial track individuals. The nodes forming the mesh self-coordinate using the unlicensed DECT NR+ 1.9 GHz band, staying away from the crowded 2.4 GHz band [8]. This method offers low interference, enabling longer-range communication and lower latency while providing higher throughput and continuous transmission capabilities [9] [10].

The proposed system delivers granular spatiotemporal data while preserving individual privacy, offering a

solution that is both extendable and easy to deploy across various indoor environments.

Preliminary results

Our conducted tests so far have focused on the optical side, by testing the Field of View (FoV) of cameras. We tested 75° and 180° FoVs; the latter (known as fisheye-view) was able to capture the entire scene in one go, which helps in reducing the overall number of deployed nodes in the proposed system. There are two approaches for people detection using the 180° FoV: (1) dewarping the frames using geometric calculations [11] [12] then feeding the frames into a chosen CNN, or (2) using a specialised CNN that can directly recognise objects in distorted images [13]. The first approach has certain trade-offs, such as losing around 30% of the FoV (as shown in Figure 1, compared to Figure 2), in addition to introducing processing overhead on the edge. As a result, the second approach has been gaining more attention in recent academic and commercial work [15] [16].

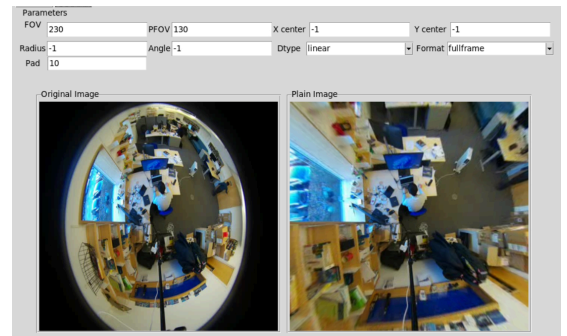


Figure 1: Dewarping 180° footage using Defisheye [12]

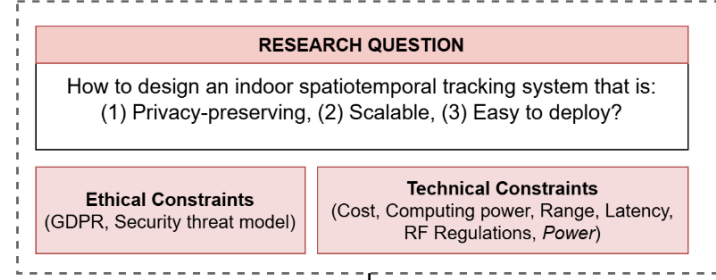


Figure 2: Attempting to detect people using the YOLOv-8n-Visdrone model [14]

Work to be done

Figure 3 shows the chosen approach for designing the proposed system based on the DSR framework [17][18]. The overall system functionality is illustrated in Figure 4. As a person enters the FoV of a sensor node

PROBLEM DEFINITION



* Based on DSR - Design Science Research framework

Figure 3: The methodology used to design the proposed system – based on DSR framework [17] [18]

(indicated by 1 on Figure 4), the node calculates a unique ID vector derived from the visual features of the individual (hair colour, worn clothes, gait shape, carried objects), which is then fused with radar's Doppler and micro-Doppler features [19] (height, distance, speed,

gait shape, body part movement patterns). The combination of those features will be used to generate an anonymous signature (feature vector) for the person. As the person moves between overlapping FoVs (2 and 3 on Figure 4), the initial node will hand over the feature vector to the neighbour node to maintain the person's ID. In the case of a person leaving a FoV and entering a non-overlapping FoV (4 and 5 on Figure 4), the system will transfer the feature vector to the next node, where the person can be re-identified locally and their trajectory reconstructed.

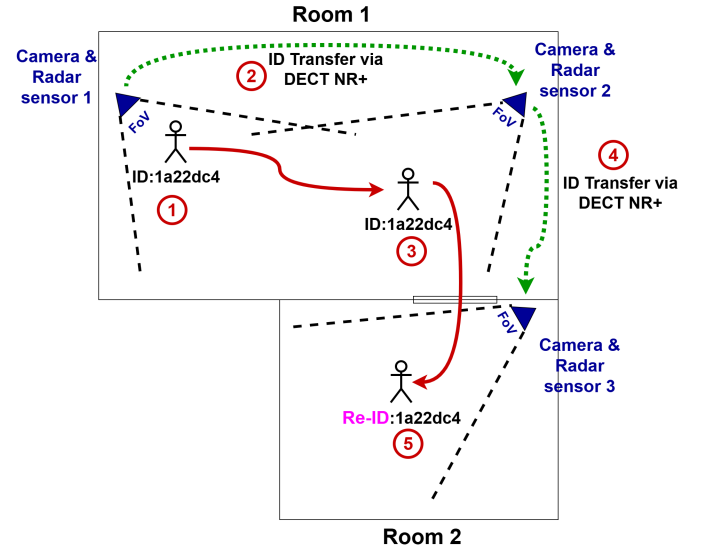


Figure 4: System functionality

The proposed system does not store camera frames or radar data; only the derived features are extracted and transmitted across the off-grid, encrypted DECT NR+ network. Furthermore, the features used to generate the temporary IDs are not stored in the system nodes and are purged once the tracking handover is complete. Ultimately, the system outputs a list of spatial coordinates paired with anonymous IDs, mapping the movement of individuals throughout a building over time.

Related work

Some existing tracking systems are based on physical tags given to individuals, such as Ubisense [20]. Although such systems usually achieve high accuracy, reaching 30 centimetres in indoor environments, they are considered intrusive, have high privacy implications, and require a complex setup. For instance, in a recent NHS trial to track patients, 50% of participants refused to carry a tag [6]. Other systems rely on traditional CCTV footage [21] to produce analytics on individuals seen across multiple cameras by running recognition software on stored footage. This approach compromises privacy and often lacks the ability to reconstruct complete spatiotemporal journeys. On the academic side, recent work [22] [23] highlights the need for more auto-calibrating and scalable sensor systems for indoor tracking.

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